

Integration Use Cases & Data Ownership

A field-by-field breakdown of what data crosses between the OFBiz-based ERP and the Hospital/Clinic Information System — including which OFBiz entities and tables are involved in each workflow.

Integration Use Cases

1 Patient Registration & Identity Sync HIS → ERP

Scenario: A new patient registers at the clinic front desk in the HIS. The ERP needs to know about them for billing purposes.

Integration: HIS pushes patient demographic data to ERP on creation/update (via FHIR Patient resource or HL7 ADT^A04).

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Full name, DOB, gender	HIS	ERP — read-only mirror
National ID / MRN	HIS	ERP — used as join key
Address, phone	HIS	ERP — needed for invoicing
Insurance info	HIS	ERP — minimal billing subset only

OFBiz tables involved: `PARTY`, `PERSON`, `PARTY_ROLE` (role "PATIENT"), `PARTY_IDENTIFICATION` (MRN/ National ID as external ID), `PARTY_CONTACT_MECH`, `CONTACT_MECH`, `POSTAL_ADDRESS`, `TELECOM_NUMBER`

2 Doctor / Provider Roster Sync HIS → ERP

Scenario: A doctor is onboarded or their role changes in the HIS. The ERP needs this for payroll, commission tracking, and order attribution.

Integration: Provider roster sync — Party + PartyRole ("Provider") pushed from HIS to ERP.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Name, specialty, license number	HIS	ERP — mirrored
Employment/contract terms, commission rate	ERP	Not present in HIS
Schedule / availability	HIS only	Not duplicated

OFBiz tables involved: `PARTY`, `PERSON`, `PARTY_ROLE` (role "PROVIDER"), `PARTY_QUALIFICATION` (license/specialty), `PARTY_RELATIONSHIP` (employment link to clinic Party), `EMPL_POSITION`, `PARTY_COMPENSATION` or custom commission entity

3 Order-to-Fulfillment (Labs, Imaging, Supplies, Procedures) Bidirectional

Scenario: A doctor orders a lab test or supply item during a consultation. ERP handles procurement, inventory deduction, and billing; HIS tracks the clinical result.

Integration: HIS sends a ServiceRequest/ORM → ERP creates an order → ERP fulfills & bills → ERP pushes status back to HIS.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Order details (item, ordering doctor)	HIS (originates)	ERP order record
Order status (created → approved → fulfilled)	ERP (owns fulfillment)	Pushed back to HIS for clinical view
Actual clinical result	HIS only	Never duplicated
Price, invoice line	ERP only	Never duplicated to HIS

OFBiz tables involved: `ORDER_HEADER` (order_type = requisition/service), `ORDER_ITEM`, `ORDER_ROLE` (linking Patient & Provider parties to the order), `ORDER_STATUS`, `STATUS_ITEM` / `STATUS_VALID_CHANGE`, `ORDER_ITEM_SHIP_GRP_INVRES` (inventory reservation), `INVENTORY_ITEM`

4 Billing & Invoicing ERP → HIS

Scenario: After a visit or order, ERP generates the invoice; HIS shows payment status at the front desk / patient portal.

Integration: ERP owns Invoice & BillingAccount; sends invoice status (paid/unpaid/partial) back to HIS.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Invoice line items, totals	ERP only	Summary shown in HIS (read-only)
Payment status	ERP	Mirrored flag in HIS
Insurance claim status	ERP / clearinghouse	Status echoed to HIS

OFBiz tables involved: `INVOICE`, `INVOICE_ITEM`, `INVOICE_STATUS`, `BILLING_ACCOUNT`, `PAYMENT`, `PAYMENT_APPLICATION`, `ORDER_PAYMENT_PREFERENCE`

5 Inventory & Facility-Level Stock (Pharmacy, Treatment Rooms) Bidirectional

Scenario: Clinic pharmacy or treatment room dispenses a medication/consumable. HIS logs clinical administration; ERP tracks stock depletion and reorder triggers.

Integration: HIS sends a dispense event → ERP deducts stock at the relevant Facility → ERP may auto-trigger a reorder.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Medication administered, dosage, clinical context	HIS only	Never duplicated
Stock quantity, lot/batch, expiry	ERP only	HIS shows "available/unavailable" flag only
Reorder thresholds	ERP only	Not in HIS

OFBiz tables involved: **FACILITY** (treatment room / pharmacy as facility or sub-facility), **FACILITY_CONTACT_MECH**, **INVENTORY_ITEM**, **INVENTORY_ITEM_DETAIL**, **PRODUCT_FACILITY** (reorder thresholds), **ORDER_HEADER** (auto-generated reorder requisition)

6 Insurance Eligibility & Claims Bidirectional

Scenario: Before a procedure, front desk checks insurance eligibility; after billing, a claim is submitted to the insurer.

Integration: HIS may check eligibility live; claim submission is triggered from ERP invoice data, referencing the HIS encounter ID.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Policy number, plan details	HIS (captured at registration)	Copied into ERP for claim generation
Encounter / visit ID	HIS	Referenced (not duplicated) in ERP
Claim status	ERP / clearinghouse	Mirrored back to HIS

OFBiz tables involved: **PARTY_ROLE** (role "INSURANCE_PROVIDER" as a Party), **AGREEMENT** / **PARTY_RELATIONSHIP** (policy link), **INVOICE**, **INVOICE_ITEM**, custom **CLAIM_STATUS** extension entity referencing **ORDER_HEADER** / **INVOICE**

7 Appointment-Triggered Financial Holds Bidirectional

Scenario: A patient books a procedure requiring prepayment/deposit; ERP needs to generate a hold or estimate before the appointment.

Integration: HIS sends appointment creation event with procedure code → ERP estimates cost, creates a pending invoice/hold → ERP confirms back whether the appointment can proceed.

DATA OWNERSHIP

FIELD	SYSTEM OF RECORD	DUPLICATED COPY
Appointment date/time, procedure type	HIS	Referenced only, not duplicated
Estimated cost, deposit status	ERP	Status flag shown in HIS scheduling screen

OFBiz tables involved: `ORDER_HEADER` (order_type = "quote/estimate"), `QUOTE` , `QUOTE_ITEM` , `ORDER_PAYMENT_PREFERENCE` (deposit/hold), `PAYMENT`

Summary: Ownership & Direction

DOMAIN	SOURCE OF TRUTH	WHAT CROSSES OVER	DIRECTION	KEY OFBIZ TABLES
Patient identity	HIS	Demographics, ID	HIS → ERP	PARTY, PERSON, PARTY_ROLE
Provider roster	HIS	Identity, role	HIS → ERP	PARTY, PARTY_QUALIFICATION
Clinical orders	HIS (origin) / ERP (fulfillment)	Order + status	Bidirectional	ORDER_HEADER, ORDER_ITEM
Billing / Invoicing	ERP	Status only	ERP → HIS	INVOICE, PAYMENT
Inventory	ERP	Stock events	Bidirectional	FACILITY, INVENTORY_ITEM
Insurance / Claims	Shared	Policy info, claim status	Bidirectional	AGREEMENT, INVOICE
Appointments / Holds	HIS (appt) / ERP (payment)	Cost estimate, hold status	Bidirectional	QUOTE, ORDER_PAYMENT_PREFERENCE

GUIDING PRINCIPLE

Never duplicate the full record — only the minimum field set needed to trigger or reflect a cross-system event. Every field must have one clearly designated system of record. This avoids the classic integration failure mode: two systems slowly drifting out of sync because both believe they own the same data.